CUSTOMER CHURN ANALYSIS

Analyze Inactive Customers Segmentation

Using SQL Reports and Segmentation

INTRODUCTION

Customer churn refers to the loss of customers who discontinue using a company's products or services. In the banking sector, customer retention is critical because acquiring new customers is generally more expensive than retaining existing ones. High churn rates can negatively impact revenue, profitability, and long-term business growth.

This project focuses on analyzing customer churn and inactive customers using SQL. The analysis was performed on a banking dataset containing information related to branches, products, customer segments, channels, revenue, customer complaints, account closures, balances, and churn rates.

Using SQL, various reports and segmentation analyses were conducted to identify churn patterns, high-risk branches, customer segments, products, and channels contributing to customer attrition. The findings help management make data-driven decisions to improve customer retention and business performance.

OBJECTIVES

The main objectives of this project are:

1. To analyze customer churn patterns in the banking sector.
2. To identify inactive customers through account closure analysis.
3. To determine branch-wise churn performance.
4. To evaluate customer segment-wise churn behavior.
5. To analyze product-wise churn rates.
6. To identify channels with the highest churn rates.
7. To study the impact of customer complaints on churn.
8. To evaluate revenue generation across branches and customer segments.
9. To identify high-risk branches and products.
10. To provide recommendations for improving customer retention and reducing churn.

DATASET DESCRIPTION

The dataset used in this project contains banking operational data related to customer activities and churn indicators.

Dataset Features:

- Date
- Branch
- Product_Type
- Channel

- Customer_Segment
- New_Accounts
- Closures
- Transactions
- Revenue
- Avg_Balance
- Customer_Complaints
- Cost_per_Account
- Total_Cost
- NPA_Percentage
- Churn_Rate

Dataset Size:

- Total Records: 38,360
- Domain: Banking
- Analysis Type: Customer Churn Analysis

The dataset provides valuable insights into customer behavior, branch performance, revenue generation, customer complaints, and churn patterns.

DATABASE DESIGN

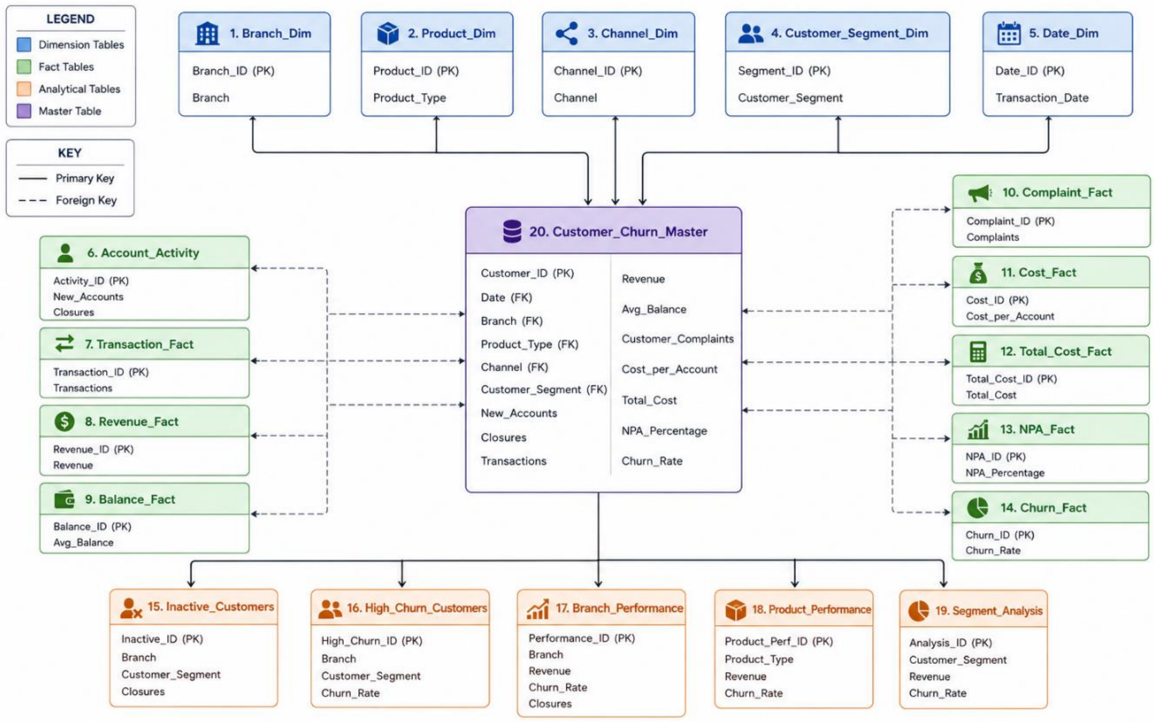
A relational database named "Customer_Churn_Analysis" was created in MySQL to store and analyze the banking dataset.

The database consists of 20 tables including:

1. Branch_Dim
2. Product_Dim
3. Channel_Dim
4. Customer_Segment_Dim
5. Date_Dim
6. Account_Activity
7. Transaction_Fact
8. Revenue_Fact
9. Balance_Fact
10. Complaint_Fact
11. Cost_Fact
12. Total_Cost_Fact
13. NPA_Fact
14. Churn_Fact
15. Inactive_Customers
16. High_Churn_Customers
17. Branch_Performance
18. Product_Performance

19. Segment_Analysis
20. Customer_Churn_Master

The main table used for analysis was "customer_churn", which contained all imported banking records.



SQL QUERY ANALYSIS

A total of 35 SQL queries were developed and executed to analyze customer churn and inactive customer behavior.

- Total records
- Revenue analysis
- Average churn rate
- Customer growth analysis
- Branch-wise churn analysis
- Segment-wise churn analysis
- Product-wise churn analysis
- Channel-wise churn analysis
- Complaint analysis
- Revenue vs churn analysis
- Balance vs churn analysis
- Closures analysis
- Above-average churn identification
- Churn classification
- Revenue ranking
- Churn ranking

- Branch performance analysis
- Segment revenue vs churn analysis
- Product performance analysis
- Executive dashboard reporting

The queries utilized SQL concepts such as:

- Aggregate Functions
- GROUP BY
- HAVING
- CASE Statements
- Subqueries
- Window Functions
- Views
- Joins

FINDINGS

The analysis produced the following key findings:

1. The overall average churn rate was 13.59%.
2. The bank achieved a positive net customer growth of 9,239 customers.
3. Tambaram Branch recorded the highest churn rate of 13.86%.
4. Retail customers experienced the highest churn rate among all customer segments.
5. Loan products showed the highest churn rate among all product categories.
6. Internet Banking recorded the highest churn rate among all service channels.
7. Tambaram Branch reported the highest number of customer account closures.
8. Customer complaints showed a relationship with churn rates across branches.
9. High-revenue branches were also experiencing relatively high churn rates.
10. Loan products generated significant revenue while showing the highest churn levels.
11. Retail customers contributed the highest revenue while also exhibiting the highest churn rate.
12. All branches demonstrated positive customer growth despite churn challenges.

CONCLUSION

Customer churn analysis plays a crucial role in improving customer retention and enhancing business performance in the banking industry. This project successfully analyzed inactive customers and churn patterns using SQL-based reports and segmentation techniques.

The findings revealed that churn was particularly high among Retail customers, Loan products, Internet Banking users, and specific branches such as Tambaram and Velachery. Customer complaints and account closures were identified as important indicators of customer attrition.

By implementing targeted retention strategies, improving customer service quality, and continuously monitoring churn trends, the bank can reduce customer loss and improve profitability.

The project demonstrates the effective use of SQL for data analysis, business reporting, customer segmentation, and decision-making support in the banking sector.